



Department of Artificial Intelligence and Machine Learning

Date: 15/04/2026	Test – I	Max. Marks : 60
Semester: VI	UG	Duration : 1.5 hrs
Course Title: Natural Language Processing and Transformers		Course Code: AI363IA

Part – A

Q. No	Question	M	B T	C O
1.1	_____ and _____ are data acquisition techniques in NLP.	2	1	1
1.2	How is raw text prepared to ensure consistency before applying NLP techniques?	2	1	1
1.3	With an example differentiate Code mixing and transliteration .	2	1	2
1.4	Analyse the different similarity measures with reference to accuracy and give example	2	2	2
1.5	Give the difference between stemming and lemmatization with an example.	2	2	2

Part – B

Q. No	Question	M	B T	C O
1	a. Elaborate Heuristic based NLP b. “Deep Learning is not the silver bullet for NLP”. Justify the given phrase	6+4	2	1
2	With a neat diagram, elaborate the NLP pipe line in detail.	10	1	1
3	a. What is Word sense disambiguation? Consider the example "apple store near me". Implement Lesk algorithm to detect disambiguation for the above example. b. An NLP model performs well on training data but fails in real-world applications with informal text, slang, and ambiguous expressions. Identify and explain the key limitations causing this performance gap.	6+4	3,2	2,1
4	A university digital library provides a search feature for students to find study materials. However, students often use different words for the same concept, causing the system to miss relevant results. (a) Design a system that expands the user query using WordNet relations, including Synonyms, Hypernyms and Hyponyms. (You can create sample document database for the same) (b) Give the advantage of using WordNet for the system	8+2	3	2
5	A job portal allows users to search for job postings using keywords. A user enters the query “developing applications”, but relevant job listings contain words like “developer”, “developed”, “application”, “apps”, etc. List and apply different types of stemmers to the words. Identify and justify the best suited stemmer in this scenario.	10	3,4	2



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Date: 23/05/2026	Test – II	Max. Marks: 60
Semester: VI	UG	Duration: 1.5 hrs
Course Title: Natural Language Processing and Transformers		Course Code: AI363IA

Part – A Scheme and Solution

Q. No	Question	M	B T	C O
1.1	_____ and _____ are two types of POS tagging.	2	1	1
1.2	"Ambiguity is one of the major challenges of PoS Tagging". Justify the statement with an example	2	2	1
1.3	Evaluate the default tagger against a tree bank corpus tagged sentences using a python code.	2	2	1
1.4	Define a transformer.	2	2	3
1.5	Give the schematic representation of attention mechanism.	2	2	2

Part – B

Q. No	Question	M	B T	C O
2	a. With a neat diagram, discuss the classifier-based tagger in detail b. With an example discuss the two-phase architecture of rule-based taggers	5+5	2	3
3	a. Construct a unigram and bigram POS tagger using NLTK. The input sentence has to be fed by the user. b. Analyze how backoff tagging improves tagging accuracy for unknown words and compare the outputs of both taggers using sample text data.	5+5	4	3
4	Develop an intelligent educational chatbot to evaluate student messages and automatically detect grammatical issues such as incomplete sentence patterns, absence of action words (verbs), and excessive usage of nouns and / or descriptive words (adjectives). The user can feed it the message and print the output.	10	3	3
5	a. With a neat diagram, elaborate the working model of ULMFiT b. Consider the code classifier = pipeline("text-classification"). Discuss the role of pipeline in detail	5+5	2	3
6	Implement the following features of transformers and discuss the steps in detail. (i) Named Entity Recognition (ii) Summarization (iii) Question Answering (iv) Text Classification. The text can be fed as an input by the user.	2.5*4= 10	3	3



Department of Artificial Intelligence and Machine Learning

Date: 18/6/2026	Test – III	Max. Marks: 60
Semester: VI	UG	Duration: 1.5 hrs
Course Title: Natural Language Processing and Transformers	Course Code: AI363IA	

Part – A

Q. No	Question	M	B T	C O
1.1	List and discuss any two challenges with Transformers.	2	1	1
1.2	_____ is used to combine the best aspects of character and word tokenization and _____ algorithm is used by hugging face for doing the same.	2	2	1
1.3	List any two prominent Encoder and Decoder transformer architectures	2	2	1
1.4	Summarize the difference between extractive vs abstractive summarization.	2	2	3
1.5	What does the following Hugging Face code return? from transformers import pipeline pipe = pipeline("text-classification", model="distilbert-base-uncased-finetuned-sst-2-english") result = pipe("This course is amazing!") print(result)	2	2	2

Part – B

Q. No	Question	M	B T	C O
2	With a neat diagram, elaborate the hugging face Eco System	2+8	2	3
3	A leading e-commerce company receives thousands of customer reviews every day for its products. Manually analyzing these reviews to determine customer satisfaction is time-consuming and inefficient. To automate this process, the company decides to develop a sentiment analysis system that classifies reviews as Positive, Negative, or Neutral. limited computational resources. As an NLP engineer, discuss the various steps involved in fine-tuning a sentiment analysis model using DistilBERT for the above scenario. Further, justify why DistilBERT is a more suitable choice for this sentiment analysis application compared to larger transformer models.	10	4	2
4	Discuss top-k and nucleus sampling techniques used in sequence generation tasks. Give the working of the same with an example.	10	3	3
5	Explain the Greedy Search decoding algorithm used in Natural Language Processing (NLP). Describe its working with a suitable example and discuss its advantages, limitations, and real-world applications.	10	2	3
6	With a neat diagram and example elaborate the role of multi head attention mechanism in encoder-decoder framework.	10	3	3